

PBL Module 1 Report Template

A Corpus-based Analysis of Latin and Greek Plural Evolutions: Predicting Standard Forms in 2040

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I . Introduction

A significant portion of modern English vocabulary — particularly the core terminology of specialized fields such as academia, science, and medicine — is rooted in Greek and Latin, and these words tend to preserve the plural forms of their source languages. However, as English transitioned into its modern period, a strong tendency toward "regularization" emerged, prioritizing user convenience over the complexity of classical plural forms. This has given rise to a linguistic conflict between the established classical plurals and the modern -s/-es endings — a tension between classical authority and modern practicality. Drawing on data from COCA and Google Ngram, this report analyzes the patterns of this shift along two axes — frequency of use and degree of specialization — and seeks to predict which forms will become standard for English speakers by 2040.

II . Research Questions

- 1) The Divergence of Modernization and Preservation: Which nouns are shifting toward regular plural forms in modern English, and conversely, which are maintaining their classical forms?
- 2) Analysis of Word Patterns by Context and Period: What specific usage patterns emerge over time and across different domains of use?
- 3) Normalization of Plural Standards: Data and media originated as the plural forms of datum and medium, respectively, yet in modern English they show a growing tendency to be used as singular nouns. Similarly, words like stigma present cases where the classical plural (stigmata) and the regularized plural

(stigmas) coexist at comparable frequencies. In light of these developments, how should the standard for plural forms be established and codified?

III. Corpus-Based Usage Analysis

Using Google Ngram Viewer, the data trends spanning approximately the past 200 years were analyzed.

The Popularization Group (words undergoing rapid regularization)

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Description of Popularization Group Graphs

- Words in the popularization group are frequently encountered in everyday contexts, and as a result, a clear trend of "regularization" is evident – that is, the replacement of complex classical plurals with the standard English rule of appending -s/-es. This reflects the principle of linguistic economy, whereby language users opt for more familiar and simpler forms in order to reduce cognitive effort.

- The Specialist Group (words retaining classical forms)

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Description of Specialist Group Graphs

- Words predominantly used within specialist communities show a strong tendency to retain their classical plural forms in order to preserve academic tradition and definitional precision. In fields such as science and medicine in particular, continuity with existing literature is paramount, and thus the pace of regularization remains very slow. Notably, however, words such as *criteria* and *data* exhibit a phenomenon of singularization — being used as singular forms even within specialist communities.

1) The Divergence of Modernization and Preservation: Which nouns are shifting to regular forms, and which retain their classical forms?

- The findings of this study confirm that patterns of plural change are closely tied to frequency of exposure — that is, how often a given word appears in everyday life.
- Words in the popularization group — those transitioning to regular forms — include *syllabus*, *appendix*, and *index*, all commonly found in general books and documents. These are rapidly shifting toward the English *-es* plural. This reflects the principle of linguistic economy, whereby the majority of language users gravitate toward forms that minimize cognitive effort.
- By contrast, words in the specialist group — those retaining classical forms — such as *bacterium*, *larva*, and *stratum*, used in scientific and medical contexts, continue to strongly preserve their classical plurals. This is due to academic conservatism within specialist communities, which prioritizes continuity with existing literature, as well as the function of classical forms as markers of domain expertise.

2) Analysis of Patterns by Context and Period: What changes emerge across time and field? (Crossover Points)

- Analysis of Ngram data for the selected nouns reveals a pattern of asymmetric regularization, in which not all words change at the same rate or in the same direction. Because resistance to linguistic change varies by context and field, the presence or absence of a crossover point — where the frequencies of the regular and classical forms intersect — serves as a decisive indicator for predicting future standards.
- The first pattern is the reversal and acceleration pattern, driven by popular diffusion. Words such as *focus*, *formula*, and *antenna*, which have migrated into everyday contexts, show on Ngram that the regular form has already overtaken the classical form or is in close competition with it. Originally confined to technical domains, these words gained momentum for change as their exposure expanded through mass media. Once a crossover occurs, the slope of the

regular form steepens — demonstrating that once a particular form surpasses the user threshold, linguistic economy drives the pace of change to accelerate exponentially.

- The second pattern is the parallel entrenchment pattern, driven by academic conservatism. Words such as bacterium, stratum, and alumnus show virtually no or negligible use of the regular form, with the two curves running parallel and never intersecting. This is because within specialist academic communities, the classical plural functions not merely as a grammatical choice, but as a sociolinguistic marker of expertise. The strong gatekeeping of specialist communities maximizes resistance to change and blocks the infiltration of regular forms. Notably, words like criteria and bacteria are undergoing semantic entrenchment, whereby the plural form itself solidifies into a collective concept, eliminating any space for regularization to occur.

3) The Restructuring of English by 2040: From Morphological Plurality to Conceptual Singularity

- Through analysis of collective nouns such as data and media, and coexistence cases such as stigma, this research team proposes the following three-stage framework for the direction standard norms should take by 2040.
- First, the recognition of collective singulars. Although data and media were originally the plural forms of datum and medium, modern English speakers perceive them not as collections of discrete elements but as unified, integrated concepts. The fact that "Data is" far outpaces "Data are" in COCA data demonstrates that users' cognitive perception takes precedence over morphological etymology. Accordingly, dictionary standards by 2040 should redefine these words not as countable plural nouns but as uncountable nouns or singular-treated nouns. Imposing the etymology of the source language undermines linguistic practicality; grammatical accommodation in line with modern intuition is therefore essential.

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- Second, the permitting of coexistence through semantic differentiation. Cases such as stigma, where the classical and regular forms coexist at comparable frequencies, are the result of semantic specialization. Stigmas has become established in general contexts referring to social stigma, while stigmata is entrenched in the specialized religious and medical context of bodily marks. This illustrates that when language changes, one form does not simply disappear; rather, both survive through functional divergence.
- Synthesizing the findings of this analysis, future language policy should adopt the following dual-track strategy:

- Learner-centered education and improvement of dictionary notation. In terms of learner-centered education, English curricula should prioritize regular forms – which carry higher cognitive efficiency – as the default standard, while collective nouns with special behavior, such as data, should be taught with singular agreement as the baseline. As for dictionary notation reform, rather than relegating classical plurals to the status of archaisms, dictionaries should clearly label them as academic or specialized-context terms, enabling users to select the appropriate variant for a given situation.

IV. Rate and Direction Comparison

The Dynamics of Asymmetric Regularization: The Correlation Between Exposure Frequency and Specialization

Drawing on the collected data, this research team analyzed three core drivers that determine the rate and direction of regularization among nouns.

1. Acceleration of Change: Frequency of Everyday Exposure (High Exposure vs. Low Exposure)

- Rapid-change group: Words frequently used in everyday education and general conversation show a sharply rising slope on Ngram curves. This reflects a strong tendency toward linguistic homogenization, whereby the majority of users resist complex inflectional changes and instead apply the basic English rule of -s/-es.
- Delayed-change group: By contrast, words such as stratum and larva – rarely encountered outside specialized textbooks – have seen the usage of their regular forms hover near zero for decades.

2. Resistance to Change: Academic Exclusivity and Specialization

The most significant variable determining the direction of change is the degree of resistance associated with the community in which a word operates.

- Resistance within specialist communities: Terminology in the fields of science, medicine, and law shows a strong tendency to reject regularization. This is because classical plural forms function as markers of professional authority within these fields. As a result, switching to a regular form is perceived within specialist communities as a sign of non-expertise, effectively suppressing the direction of change.

3. The Inflection Point of Change: Semantic Restructuring

The most distinctive directional pattern is exhibited by data and criteria. Rather than shifting toward regular forms (datas, criterions), these words are moving toward singularization – a process in which the plural form itself is absorbed into

singular usage.

- This suggests that users' cognitive shift – perceiving not individual data points but the entire data set as a single unified concept – is operating more powerfully than morphological rules.

V. Predictions for 2040

Synthesizing the data derived from corpus analysis with findings from prior linguistic research, this research team projects the landscape of English plural standards in 2040 through the following three scenarios.

First, the completion of regularization for everyday vocabulary. Words with high everyday exposure, such as *focus* and *formula*, will see their regular plural forms established as the virtually exclusive standard by around 2040. In accordance with the principle of grammatical economy proposed by Lee Eun-kyung (2001), language users come to prefer the efficient *-s/-es* forms over complex classical rules in order to reduce cognitive load. Having already passed their crossover points on Ngram and tracing steep upward curves, these words are expected to fully displace their classical forms in general discourse by 2040, with the classical forms becoming restricted to specialized, rather than everyday, usage.

Second, the entrenchment and identity preservation of academic and technical terminology. Words primarily used in specialized academic contexts – such as *bacterium* and *stratum* – will retain the dominance of their classical plural forms well into 2040. According to Lee Sang-do and Choi Young-i (2019), unlike lexical borrowing, grammatical structures tend to be resistant to change. In specialist fields in particular, classical plural forms serve as a kind of identity marker for the relevant academic community, and as a result, these words are predicted to maintain high resistance to regularization and remain outside its reach even in 2040.

Third, the semantic differentiation of contested words and the establishment of dual standards. Words currently in close competition between two forms – such as *stigma* – are expected not to converge on a single form, but rather to present a model of fully realized semantic differentiation. As demonstrated by Moon Sang-ho (2015) through big data analysis, shifts in keyword frequency are closely tied to context of use. Accordingly, by 2040, *stigma* is projected to split entirely into two registers: the everyday context of social stigma (*stigmas*) and the specialized context of religious and physical marks (*stigmata*) – establishing a flexible dual-standard system in which users select the optimal variant according

to the register in which they are operating.

VI. Recommendations for Dictionaries and Education Policy

1. For Lexicographers

- Frequency-based priority listings: Words for which regularization has accelerated due to high everyday exposure should have their regular forms elevated to the primary headword. Classical plural forms, rather than being dismissed as archaisms, should be clearly labeled as academic or specialized-context terms, guiding users to select the appropriate variant for a given situation.
- Reclassification of collective singulars: Words that modern users perceive as a single unified concept – such as data and media – should be redefined in dictionary standards not as countable plural nouns but as uncountable nouns or singular-treated nouns.
- Permitting coexistence through semantic differentiation: In cases where meaning has become contextually entrenched – such as stigma (stigmas vs. stigmata) – dictionaries should list each form alongside its respective domain (general social stigma vs. religious marks), thereby supporting the functional divergence of the language.

2. For Curriculum Designers

- Learner-centered, staged curriculum design:
 - > Primary and secondary education: Prioritize regular forms (-s/-es) as the standard to reduce the cognitive burden on learners.
 - > Higher and professional education: For fields requiring academic precision (such as science and medicine), teach the historical context of classical plural forms and their function as markers of professional identity as a selective alternative.
- Grammar instruction reflecting modern intuition: Rather than imposing etymological rules, instruction should prioritize the practicality of modern English. It is recommended that special collective nouns such as data be taught with singular agreement as the default.
- Context-centered assessment of appropriateness: Educational guidelines should be revised to approach plural form selection not as a binary right-or-wrong question, but from the perspective of appropriateness relative to audience and context – distinguishing between everyday and specialist registers.

VII. Conclusion

This study combined statistical data from Google Ngram and COCA with linguistic theory to analyze the plural standards toward which nouns of Latin and Greek origin will converge by 2040. The findings confirm that change in English plural

forms is not moving in a single direction, but rather forming a bifurcated structure that diverges along two axes: frequency of exposure and degree of specialization.

For words with high everyday exposure, Ngram data already shows crossover points at which regular forms have overtaken classical forms, and by 2040 the regular form will have settled as the virtually exclusive standard. This is the outcome of users minimizing cognitive load in accordance with the principle of grammatical economy, while classical forms are expected to disappear from everyday discourse and survive only in restricted, specialized written contexts.

For words confined to academic and specialist contexts, classical plural forms will retain their dominant status well into 2040. In these words, the classical form functions not merely as a choice but as a marker of professional identity within the relevant academic community. The parallel entrenchment pattern observed on Ngram demonstrates that the normative gatekeeping of specialist communities is maximizing resistance to linguistic change.

Words undergoing semantic differentiation — such as stigma — are expected to settle not into a single unified form, but into a dual-standard system calibrated to context of use.

Words such as data, media, and criteria are following a distinct trajectory — not a transition to regular forms, but a recategorization of the plural form itself as an uncountable collective noun. The overwhelming dominance of "Data is" over "Data are" in COCA data suggests that users' cognitive restructuring is operating more powerfully than morphological origin, and this represents a key shift that must be reflected in future language policy.

In sum, the plural standards of English in 2040 will not constitute a single unified norm, but a flexible, multilayered normative system. In response to these changes, this study recommends the revision of lexicographic standards and a phased reform of educational curricula, and emphasizes that policy wisdom capable of maintaining the balance between linguistic practicality and academic tradition will be an enduring necessity.

VIII. References

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